



Hah Min Lew

Ph.D. student · Korea University

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Biography

Ph.D. student at the Visual & General Intelligence (VGI) Lab., Korea University, supervised by Prof. Gyeong-Moon Park. Current research focus lies in Vision-Language-Action (VLA) models, Vision-Language-Navigation (VLN) models, and multimodal generative models.

Recipient of the [Presidential Science Scholarship](#) funded by the Ministry of Science and ICT.

Prior to academia, professional experience includes over three years as an AI Researcher at Klleon AI Research. Key contributions involved photo-realistic 3D human generation, enabling lifelike avatar movements and lip-sync. Holds both M.S. and B.E. degrees from DGIST, with a strong foundation in Deep Learning and Signal Processing, advised by Prof. Jae Youn Hwang.

Education

Korea University (Advisor: Prof. Gyeong-Moon Park)	Seoul, South Korea
Ph.D. in Artificial Intelligence	Mar. 2026 - present
DGIST (Advisor: Prof. Jae Youn Hwang)	Daegu, South Korea
M.S. in Electrical Engineering and Computer Science	Mar. 2019 - Aug. 2022
DGIST (Daegu Gyeongbuk Institute of Science and Technology)	Daegu, South Korea
B.E. in School of Undergraduate Studies	Mar. 2014 - Feb. 2019

Experience

Klleon AI Research	Seoul, South Korea
AI Researcher	Aug. 2022 - Feb. 2026
- Accelerated inference using a training-free diffusion sampler (4.11×). - Developed audio-driven 3DMM generation for virtual avatars with natural human-like expressions and movements. (LVE 27.5%, FDD 28.9%, MEE 27.1%, CE 24.1%, Diversity 17.7%) - Developed a photorealistic head rendering model using Gaussian Splatting, outperforming 5 state-of-the-art models. (MSE 59.96%, PSNR 4.41dB, SSIM 3.85%, LPIPS 38.16%) - Developed a Head Swap AI model with 212.7x speedup, reducing annual GPU costs by 99.53% (from \$2.2M to \$10.5K). - Built large-scale multimodal data pipelines (9.41M+ frames from in-the-wild videos). - Integrated external APIs (NVIDIA Omniverse Audio2Face, OpenAI ChatGPT, TTS) into a streaming avatar system, achieving 25-28 FPS performance.	
LANTERN	Daegu, South Korea
Co-founder	Nov. 2016 - Jul. 2017
- Founded a data-driven personalized tutor matching service company in collaboration with Class101. - Designed a matching database system and established tutor evaluation metrics for personalized recommendations.	

Publications

[P1] FacePlex: Toward Natural Full-Duplex Conversational Avatars	First Author
H. Lim*, Hah Min Lew* , J.-H. Lee*, M.-J. Kim, S. Lee, J.-S. Kang, and G.-M. Park. arXiv 2026	Sep. 2026
[C5] AudioAvatar: Personalized Audio-driven Whole-body Talking Avatars	Co-Author
S. Lee, S. Moon, Hah Min Lew , J.-S. Kang, and G.-M. Park. CVPR 2026	Jun. 2026
[C4] SMAGA: Secondary Motion-Aware 3D Clothed Gaussian Avatars from Monocular Videos	Co-Author
S. Lee, S. Moon, Hah Min Lew , J.-S. Kang, and G.-M. Park. ICLR 2026	Apr. 2026
[C3] GeoAvatar: Adaptive Geometrical Gaussian Splatting for 3D Head Avatar	First Author
S. Moon*, Hah Min Lew* , S. Lee, J.-S. Kang, and G.-M. Park. ICCV 2025	Oct. 2025
[C2] Towards High-fidelity Head Blending with Chroma Keying for Industrial Applications	First Author
Hah Min Lew* , S.-M. Yoo*, H. Kang*, and G.-M. Park. WACV 2025	Mar. 2025
[C1] CSS-Net: Classification and Substitution for Segmentation of Rotator Cuff Tear	Co-Author
K. Lee, Hah Min Lew , M. H. Lee, M. Kang, J. Kim, and J. Y. Hwang. ACCV 2022	Dec. 2022
[J6] Multi-task and Few-shot Learning-based Fully Automatic Deep Learning Platform for Mobile Diagnosis of Skin Diseases	Co-Author
K. Lee, T. C. Cavalcanti, S. Kim, Hah Min Lew , D. H. Suh, D. H. Lee, and J. Y. Hwang. IEEE JBHI (IF: 7.021)	Jan. 2023

[J5] Deep Learning-based Framework for Fast and Accurate Acoustic Hologram Generation M. H. Lee, Hah Min Lew , S. Youn, T. Kim, and J. Y. Hwang. IEEE TUFFC (IF: 3.267)	<i>Co-Author</i> Dec. 2022
[J4] Speckle Reduction via Deep Content-Aware Image Prior for Precise Breast Tumor Segmentation in an Ultrasound Image H. Lee, M. H. Lee, S. Youn, K. Lee, Hah Min Lew , and J. Y. Hwang. IEEE TUFFC (IF: 3.267)	<i>Co-Author</i> Sep. 2022
[J3] Intelligent Smartphone-based Multimode Imaging Otoscope for the Mobile Diagnosis of Otitis Media T. C. Cavalcanti, Hah Min Lew , K. Lee, S. Lee, M. K. Park, and J. Y. Hwang. Biomedical Optics Express (IF: 3.562)	<i>Co-Author</i> Dec. 2021
[J2] Ultrasonic Blood Flowmeter with a Novel Xero Algorithm for a Mechanical Circulatory Support System Hah Min Lew , H. Shin, M. H. Lee, S. Youn, H. C. Kim, and J. Y. Hwang. Ultrasonics (IF: 4.062)	<i>First Author</i> Aug. 2021
[J1] Forward-Looking Multimodal Endoscopic System Based on Optical Multispectral and High-Frequency Ultrasound Imaging Techniques for Tumor Detection J. Kim, Hah Min Lew , J. Kim, S. Youn, H. A. Faruque, A. N. Seo, S. Y. Park, J. H. Chang, E. Kim, and J. Y. Hwang. IEEE TMI (IF: 11.037)	<i>Co-Author</i> Feb. 2021

Patents

COMPUTING DEVICE FOR HEAD SWAPPING Hah Min Lew , H. Kang, S.-M.Yoo, G.-M.Park (WO2025042068A1, KR1020230154188A)	<i>Application</i> Feb. 2025
MOBILE OTOSCOPE SYSTEM J. Y. Hwang, T. C. Cavalcanti, Hah Min Lew (KR102648059B1)	<i>Patent</i> Mar. 2024
ULTRASONIC BLOOD FLOW MEASURING APPARATUS AND METHOD THEREOF J. Y. Hwang, Hah Min Lew , H. C. Kim (KR102514633B1)	<i>Patent</i> Mar. 2023
BLADDER MONITORING APPARATUS AND METHOD FOR CONTROLLING BLADDER MONITORING APPARATUS J. Y. Hwang, M. H. Lee, Hah Min Lew (KR102460829B1)	<i>Patent</i> Oct. 2022
THREE-DIMENSIONAL DIAGNOSTIC SYSTEM J. Y. Hwang, J. Kim, Hah Min Lew (KR102379481B1)	<i>Patent</i> Mar. 2022

Awards & Scholarships

Presidential Science Scholarship for Graduate Students (3rd Cohort) Ministry of Science and ICT	<i>South Korea</i> Mar. 2026 – Present
Outstanding Poster Award 2021 Student Conference at DGIST	<i>Daegu, South Korea</i> 2021
Best Paper Award 2021 Spring Conference at KOSOMBE	<i>Remote, South Korea</i> 2021
Best Project Award 2016 Undergraduate Group Research Project (UGRP) Program at DGIST	<i>Daegu, South Korea</i> 2017
Full Government Scholarships Full tuition exemptions and school expenses support	<i>Daegu, South Korea</i> 2014-2022

Skills

Programming	Python, Bash, MATLAB, C, Java
Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn, Pytorch3D
Tools	Docker, Containerd, Git, Node.js, HTML, CSS, JavaScript
Languages	Korean, English